

AI-Based Learning & Assessment System

1. Daily Learning – AI Video Generation

Concept:

This feature uses Artificial Intelligence to automatically generate daily learning videos for trainees without human recording.

Process:

- AI selects the “Topic of the Day” based on trainee progress.
- AI generates a learning script (technical or soft skills).
- The script is converted into voice using Text-to-Speech.
- AI generates a video using avatar or animation tools.

Output:

A high-quality learning video is delivered to the trainee every morning automatically.

Automation:

The system dynamically generates content based on each trainee’s learning progress, ensuring personalized learning.

2. Weekly Assessment – AI Generation & Validation

Concept:

This feature automatically generates weekly assessments and evaluates trainee performance using AI without manual intervention.

Process:

- AI generates questions based on the content learned during the week.
- Questions include MCQs, True/False, and subjective questions.
- Trainees submit their answers through the system.

Validation (RAG - Retrieval Augmented Generation):

AI validates answers by comparing them with stored learning content, ensuring accurate evaluation based on actual study material.

Scoring:

- Objective questions are auto-scored instantly.
- Subjective answers are evaluated using AI based on logic and understanding.

Result:

The system provides instant results and feedback without any manual grading effort.

3. Technologies Used

- AI Engine: OpenAI (GPT models)
- Voice Generation: Text-to-Speech APIs
- Video Generation: AI Avatar Tools (D-ID, Synthesia)
- Backend: Node.js
- Frontend: React
- Storage: Cloud (AWS S3 / Database)

4. Key Benefits

- Fully automated learning and assessment system
- No manual content creation or grading required
- Personalized learning experience
- Scalable and efficient solution
- Consistent and high-quality output